

- 6 (a) (i) 10.0 eV electron has sufficient energy to excite mercury atom from ground state to energy level of  $-1.57$  eV (because  $-1.57 - (-10.38) = 8.81$  eV).

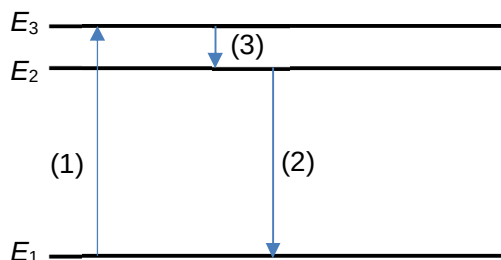
As atom de-excites from  $-1.57$  to  $-5.74$  eV, a photon is emitted. [1]

$$\frac{hc}{\lambda} = [-1.57 - (-5.74)] \times (1.6 \times 10^{-19})$$

$$\lambda = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{6.672 \times 10^{-19}}$$

$$\lambda = 2.98 \times 10^{-7} \text{ m} \quad [1]$$

(ii)



- (iii) The phosphor atom absorbs a single high-energy UV photon, but it de-excites in multiple, smaller steps (or 2 steps). (Hence the emitted photons have less energy and longer wavelengths.) [1]

- (b) (i) An incident photon transfers its entire energy (, given by  $E = hf$ ,) to a single electron (on the metal surface). (note: emphasis is one-to-one, cannot accept photons) [1]

Part of this energy is used by the electron to overcome the work function of the metal. The remaining energy is converted into the (maximum) kinetic energy of the emitted photoelectron. [1]

- (ii) Photon energy:  $E = hc / \lambda = (6.63 \times 10^{-34})(3.00 \times 10^8) / (310 \times 10^{-9}) = 6.42 \times 10^{-19} \text{ J}$   
 Work function:  $\Phi = 2.28 \times 1.60 \times 10^{-19} = 3.65 \times 10^{-19} \text{ J}$  [1 for E or  $\Phi$ ]  
 $KE_{\max} = E - \Phi = (6.42 \times 10^{-19}) - (3.65 \times 10^{-19}) = 2.77 \times 10^{-19} \text{ J}$  [1]

- (c) (i)  $KE = p^2/2m$   
 $p = \sqrt{2(9.11 \times 10^{-31})(2.77 \times 10^{-19})} = 7.10 \times 10^{-25} \text{ kg m s}^{-1}$  [1]

de Broglie wavelength:  $\lambda = h/p = 6.63 \times 10^{-34} / (7.10 \times 10^{-25}) = 9.34 \times 10^{-10} \text{ m}$ . [1]

- (ii) A pattern is formed because the electron's wavelength is of the same order of magnitude as the atomic spacing of about  $10^{-10} \text{ m}$ . [1]

- (iii) Electrons with lower kinetic energy have longer wavelengths and are diffracted more (or range of wavelengths and range of angles of diffraction) [1]

This overlapping of many patterns causes the observed pattern to be smeared or blurred (sharp inner ring due to electrons with max KE). [1]

